tf.compat.v1.nn.sampled_softmax_loss Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/nn/sampled_softmax_loss

> *Computes and returns the sampled softmax training loss.*

Function Signatures

```
tf.compat.v1.nn.sampled_softmax_loss( weights, biases,
labels, inputs, num_sampled, num_classes, num_true=1,
sampled_values=None, remove_accidental_hits=True,
partition_strategy='mod', name='sampled_softmax_loss',
seed=None )
```

Code Examples

```
tf.compat.v1.nn.sampled_softmax_loss(
weights,
biases,
labels,
inputs,
num_sampled,
num_classes,
num_true=1,
sampled_values=None,
remove_accidental_hits=True,
partition_strategy='mod',
name='sampled_softmax_loss',
seed=None
)
```


```
tf.compat.v1.nn.sampled_softmax_loss(
weights,
biases,
labels,
inputs,
num_sampled,
num_classes,
num_true=1,
sampled_values=None,
remove_accidental_hits=True,
partition_strategy='mod',
name='sampled_softmax_loss',
seed=None
)
```

```
if mode == "train":
loss = tf.nn.sampled_softmax_loss(
weights=weights,
biases=biases,
labels=labels,
inputs=inputs,
...,
partition_strategy="div")
elif mode == "eval":
logits = tf.matmul(inputs, tf.transpose(weights))
logits = tf.nn.bias_add(logits, biases)
labels_one_hot = tf.one_hot(labels, n_classes)
loss = tf.nn.softmax_cross_entropy_with_logits(
labels=labels_one_hot,
logits=logits)
```


```
if mode == "train":
loss = tf.nn.sampled_softmax_loss(
weights=weights,
biases=biases,
labels=labels,
inputs=inputs,
...,
partition_strategy="div")
elif mode == "eval":
logits = tf.matmul(inputs, tf.transpose(weights))
logits = tf.nn.bias_add(logits, biases)
labels_one_hot = tf.one_hot(labels, n_classes)
loss = tf.nn.softmax_cross_entropy_with_logits(
labels=labels_one_hot,
logits=logits)
```

Documentation

Computes and returns the sampled softmax training loss.

This is a faster way to train a softmax classifier over a huge number of classes.

This operation is for training only. It is generally an underestimate of the full softmax loss.

A common use case is to use this method for training, and calculate the full softmax loss for evaluation or inference. In this case, you must set partition_strategy="div" for the two losses to be consistent, as in the following example:

See our Candidate Sampling Algorithms Reference
(pdf).
Also see Section 3 of (Jean et al., 2014) for the math.

Returns

References